Konstantinos Gaitanis

Mobile Architect & Principal iOS Engineer

Zürich, Switzerland

Portfolio: kgaitanis.github.io
k.gaitanis@protonmail.com
+41 76 572 81 89
linkedin.com/in/gaitanis
github.com/kgaitanis

Profile

Mobile architect and principal iOS engineer with 13+ years of experience building mobile products. I've built applications as iOS and Android have evolved, learning how these platforms work, why they've changed and where their limits are. Today, I use AI tools to work faster. My experience allows me to question their output, catch mistakes and make sound engineering decisions.

I specialise in banking, fintech and blockchain applications, with a focus on security. I've led mobile development teams in Switzerland, working on a crypto exchange app and a shared platform used by approximately 25 banks. I also created and maintain IcpKit, an open-source Swift SDK for the Internet Computer blockchain, used in production by Tangem and Bity.

Experience

Bity SA · Neuchâtel, Switzerland

2022 - Mar 2026

Mobile Architect (iOS & Android) · Principal iOS Engineer

- Built a crypto wallet and exchange app from the ground up, including crypto transactions, KYC and credit card payments.
- Designed the architecture for both apps and implemented the native iOS app in Swift and SwiftUI.
- Separated the UI from the native business logic, wallet, blockchain and security components. This allowed us to migrate from Flutter to SwiftUI on iOS and Jetpack Compose on Android without rewriting the core logic.
- Implemented self-custody and secure private-key management.
- Integrated Bitcoin, Internet Computer and EVM networks, including Ethereum and Arbitrum, with support for tokens and NFTs.
- Built production and test implementations of each service, supporting deterministic tests and offline SwiftUI previews without a live backend. Managed GitLab CI/CD, App Store releases and production issues.

IcpKit

2022 - Present

Creator & Principal Developer

GitHub · Swift Package Index

- Creator and maintainer of IcpKit, an open-source Swift SDK for communicating with the Internet Computer blockchain.
- Implemented cryptographic signing, digital identities, blockchain communication, serialisation and Swift code generation.
- Used in production by Bity and Tangem, with development supported by a grant from the DFINITY Foundation.

CREALOGIX AG · Zürich, Switzerland

2020 - 2022

Mobile Software Architect

- Designed and maintained a shared mobile platform used by approximately 25 Swiss banks, including cantonal, retail and private banks.
- Built a configuration-driven architecture allowing each bank to have its own branding, backend integrations, security requirements and optional features, while keeping the core application shared.
- Developed pipelines to automate building, signing, hardening and distribution.

Core Expertise

- Mobile architecture, technical strategy and hands-on development
- Native iOS (Swift, SwiftUI, UIKit, Swift Concurrency)
- Cross-platform development (Flutter, Kotlin/Android, web-based apps)
- Banking, fintech, blockchain and mobile security
- Cryptography, self-custody wallets, digital identities and blockchain protocols
- Shared mobile platforms, modular architecture and code reuse
- Dependency injection, automated testing and CI/CD
- App Store releases, deployment and production maintenance
- Technical leadership, team management, hiring and mentoring

Technology & Protocols

- Swift 6, SwiftUI, UIKit, Objective-C, Swift Package Manager
- Flutter
- GitLab CI/CD, Jenkins, Groovy, Promon
- Cryptography, Candid, CBOR
- Bitcoin, Ethereum/EVM, Internet Computer, ERC-20, ERC-721, ICRC-1/2/3/7
- Core Data, WebSockets, KYC, NFC
- C++, Qt/QML, OpenGL

Konstantinos Gaitanis **Mobile Architect & Principal iOS Engineer** kgaitanis.github.io

CREALOGIX AG · Zürich, Switzerland
2019 - 2020

Mobile Tech Lead

- Led the mobile development team, setting the technical direction and making architectural decisions across multiple banking projects.
- Reviewed code, established development standards and helped engineers solve complex technical problems.
- Coordinated work on the shared platform and releases for individual banking apps.
- Conducted technical interviews for hiring new engineers.

Numbrs Personal Finance AG · Zürich, Switzerland
2018 - 2019

Engineering Manager, Payments

- Led a team of eight engineers across iOS, Android, backend and QA, responsible for payments and transaction management.
- Managed hiring, mentoring and performance reviews, and guided the team's technical decisions.

Senior iOS Engineer 2017 - 2018

- Developed banking features for payments, transfers, recurring orders and transaction history using Swift, UIKit and Objective-C.
- Worked on security-sensitive financial transactions and integration with banking backend services.

eyeMaps - Augmented Reality Maps
2013 - 2017

Founder & Principal Engineer eyemaps.org

- Created eyeMaps, an augmented-reality map that overlaid geographic features on the landscape as seen by the user.
- Combined GPS, accelerometer, gyroscope, magnetometer, digital elevation data, and OpenStreetMap features within a local 3D coordinate system.
- Built the rendering engine, offline map storage, and backend infrastructure in C++, OpenGL, GLSL and Qt/QML.
- Shared rendering and geographic-data layers while keeping platform integrations native.
- Synchronized the native UI and real-time 3D model at 60 fps across iOS and Android.

iMagX · IBA, Belgium
2010 - 2013

C++ Developer

- Designed and implemented user interfaces for proton-beam cancer treatment applications using C++ and Qt.
- Developed 3D visualization tools for high-bit-depth medical imaging data.

UCL - TELE Laboratory · Belgium
2008 - 2010

Research Assistant

- Developed machine-learning and optimization methods for selecting compression parameters from subjective video-quality assessments.

Education

Université catholique de Louvain
Louvain-la-Neuve, Belgium

- **2004-2008 · PhD**
  Engineering Sciences
  Thesis: Team Behavior Recognition using Dynamic Bayesian Networks
- **2001-2004 · Master**
  Electromechanical Engineering, Mechatronics
- **1999-2001 · Bachelor**
  Engineering Sciences

Languages

- **English C2**
  Cambridge Certificate of Proficiency
- **French C2**
  Diplôme d'études françaises, Sorbonne II
- **Greek**
  Native
- **Spanish**
  Conversational